

First week

A social science a government made calculation (managed allocated their resources) unlimited wants corresponding limited resources.

→ There is no economy if the resources are infinite

Free goods: The goods that are available without any effort or money.

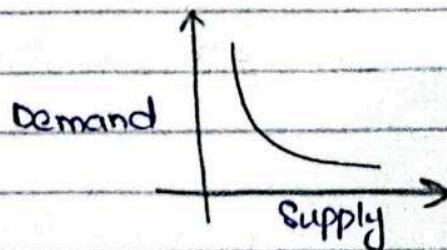
Non-Free goods: The goods that are available only with efforts or money.

Demand and Supply:

If demand is high & supply is low, the price will be high and vice versa.

* Both are inversely proportional to each other and effect the price.

* We can evaluate a project that will be successful in future can be done by different analysis.



Factors of Production:

There are three factors:

- ① LAND ② LABOUR ③ CAPITAL STOCK

Interest Rate & Rate of Return

- Time value of Money: It changes with time.
Interest rate is the percentage of interest amount. It will be in percentage & interest will be in dollars.

$I \rightarrow$ Interest (\$, R)

$i \rightarrow$ Interest Rate (%)

$\therefore$ you must write the per unit with the answer of interest rate with %.

Terminology and Symbols

$P \rightarrow$ value of money at present time or at time 0

$F \rightarrow$ value of money at some time in Future

$A \rightarrow$ series of consecutive, equal, end, amount of money

$n \rightarrow$ Number of interest period

$i \rightarrow$ Interest rate per time period

$t \rightarrow$ Time, stated in periods; years, months, days

Example 1.3

An employee borrows \$10,000 on May 1, must repay a total of \$10,700 exactly one year later.

Interest Amount and Interest Rate Paid

1) Interest = amount owed now - principle

2) Interest Rate (%) = $\frac{\text{interest occurred per time}}{\text{principle}} \times 100\%$

1) Interest paid = $\$10,700 - \$10,000 = \$700$

2) Interest rate (%) = $\frac{\$700}{\$10,000} \times 100\% = \boxed{7\% \text{ per year}}$

Example 1.4

Plans to Borrow \$20,000 from a bank for

1 year at 9% interest.

a) compute the interest & total amount due after 1 year

b) Total amount after 1 year, loan interest, rate of 9%.

Total Due = principle (1 + interest rate)

→ From the perspective of a saver, a lender or an investor → Interest Earned

Interest Earned = Total amount now - principle

→ Interest earned over a specific period of time is expressed as a percentage of the original amount and is called rate of Return (RoR)

$$\text{Rate of return (\%)} = \frac{\text{interest occurred per unit time}}{\text{Principal}} \times 100\%$$

Solution:

$$\begin{aligned} \text{Total Due Amount} &= \$20,000 (1 + 9\%) \\ &= \boxed{\$21,800} \end{aligned}$$

Example #1.5

- Calculate the amount deposited 1 year ago to have \$1000 now at an interest rate of 5% per year.
- Calculate amount of interest earned during this time period.

$$\begin{aligned} \text{Total occurred} &= \text{Deposit} + \text{Deposit (interest Rate)} \\ \text{Amount owed} &= \$10,000 \\ \text{Interest rate} &= 5\% \end{aligned}$$

$$\begin{aligned} \text{Total amount occurred} &= \text{Principal} (1 + i) \\ \$1000 &= \text{Principal} (1 + 5\%) \\ \frac{\$1000}{1.05} &= \text{Principal} \end{aligned}$$

$$\text{Principal} = \$952.38$$

Amount of interest Earned

$$\text{Interest Earned} = \text{Total amount now} - \text{Principal}$$

$$\text{Interest Earned} = \$1000 - \$952.38$$

$$= \boxed{\$47.62}$$

Example 1.6

Borrowed \$5000. Two ways loan repay
Engineering Economics symbol?

- a) Five equal annual installments with interest based on 5% per year

$$P = \$5000, i = 5\% \text{ per year}$$

$$n = 5 \text{ years}, A = ?$$

P = value of money at a time designated
Present worth (W)

i = Interest Rate per time period

n = Number of interest periods

A = Annual worth A(W)

- b) One Payment 3 years from now with interest based 7% per year

$$P = \$5000, i = 7\%, n = 3 \text{ years}, F = ?$$

F = value of money at some future time

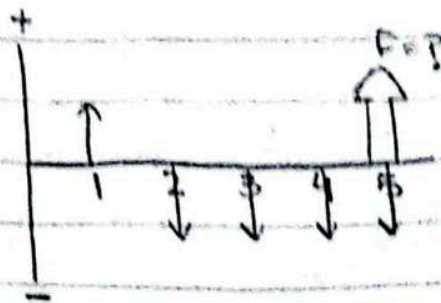
Example 1.7

Statement From Book - Data is given below

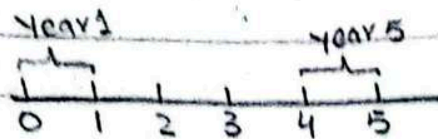
$$P = \$5000, i = 6\% \text{ per year}, A = \$1000 \text{ for years}$$

$$n = 5, F = \text{at end of 6th year}$$

- CASH FLOWS: ESTIMATION & DIAGRAMMING
Net cash flow = cash inflows - cash outflows



example of +ve and -ve cash flows



A typical cash-flow scale of 5 years

+ve axis: Getting money (withdrawal)

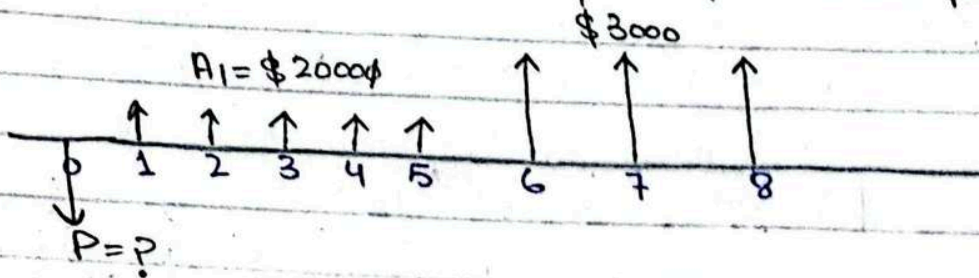
-ve axis: Paying money (deposit)

length of line shows amount of money

Example 1.10

Data: $P=?$, $A_1 = \$2000$, $A_2 = \$3000$

$n_1 = 5$ years, $n_2 = 3$ years, $i = 8.5\%$ per year



(3)

Simple and Compound Interest

- Based on principal amount is simple interest
- calculated on principal amount & outstanding amounts leads to compound interest.

* Simple interest = (Principal)(n)(i)

* Compound interest = (Principal + all accurate interest)(interest rate)

Example 1.14

Lended amount = \$100,000; 3 years loan
10% per year; money repay end of 3 years

$$\text{Interest per year} = (100,000)(10\%) \\ = 10000$$

$$\text{Total interest} = (100,000)(3)(0.10) \\ = 30000$$

$$\text{Total due} = 100,000 + 30000 = \$130000$$

Now after 3 years

$$\text{Interest per year} = (100,000)(0.10) = 10000$$

$$1 \text{ year} = 100000 + 10000 = 110000$$

$$2 \text{ year} = 110,000 + 11000 = 121,000$$

$$3 \text{ year} = (121)(0.10) = 12,100$$

$$121,000 + 12,100 = \$133,100$$

Table 1.1 Repay \$5000 in 5 years at 8% year

(simple interest)

End of year	Interest owed for year	Total owned at end of year	End of year payment	Total owned after payment
0		\$5000		
1	400	\$5400	0	\$5400
2	400	\$5800	0	\$5800
3	400	\$6200	0	\$6200
4	400	\$6600	0	\$6600
5	400	\$7000	\$7000	0

b) Repay \$5000 in 5 years at 8% (Compound)

0		\$5000		
1	400	\$5400	0	\$5400
2	432 $5400 \times \frac{8}{100}$	\$5832 $5400 + 432$	0	\$5832
3	466.56	\$6298.56	0	\$6298.56
4	503.88	\$6802.44	0	\$6802.44
5	544.19	\$7390.19 \$7346.63	\$7346.63	0

c) Pay annually (Compound)

0		\$5000		
1	400	\$5400	400	\$5000
2	400	\$5400	400	\$5000
3	400	\$5400	400	\$5000
4	400	\$5400	400	\$5000
5	400	\$5400	5400	0

d) 1/5th of principal amount annually = 1000

0		\$5000		
1	400	\$5400	\$1000	\$4000
2	320	\$4320	\$1320	\$3000
3	240	\$3240	\$1160 \$1240	\$2000
4	160	\$2160	\$1080 \$1160	\$1000
5	80	\$1080	\$1080	0

$$\begin{aligned}
 * A &= P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] \\
 &= 5000 \left[\frac{0.08(1+0.08)^5}{(1+0.08)^5 - 1} \right] \\
 &= \$1252.28
 \end{aligned}$$

0		\$5000		
1	400	\$5400	\$1252.28	\$4147.72
2	331.81	4479.53	1252.28	\$3227.25
3	258.18	3485.43	1252.28	\$2233.15
4	178.65	2411.802	1252.28	\$1159.52
5	92.76	1252.28	1252.28	0

(4)

Minimum Attractive Rate of Return (MARR)

Financing
 ↳ Equity ↳ Debt

Chapter no 2

find
given

Single amount factor (P/F, F/P)

$$F = P(1+i)^n \text{ or } P = \frac{F}{(1+i)^n}$$

P has value than F
less

↓
1000

5
↓
F

there are three methods to solve these questions

- ↳ Using tables
- ↳ Using formula
- ↳ Using MS Excel

Format to attempt the question :-
Four Steps

- 1- Draw CashFlow diagram
- 2- Write equation i.e. $F = P(F/P, i, n)$
- 3- Write formula
- 4- Put values and write answer

Example 2.1

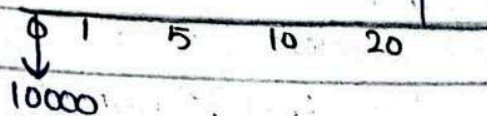
Bonus = \$10,000 ; Profit = 8% per year ,

After 20 years

$P = \$10,000$, $F = p$, $i = 8\%$, $n = 20$ years

$i = 8\%$

$F = p$



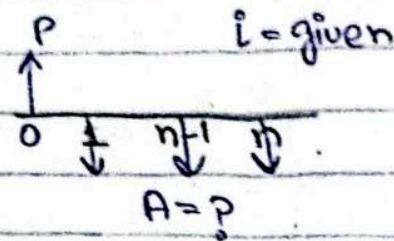
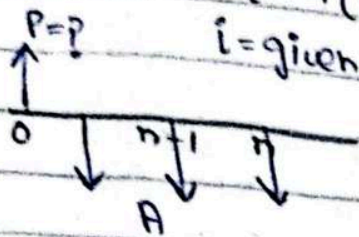
$$\begin{aligned} \rightarrow F &= P(F/P, i, n) \\ &= 10000(F/P, 0.08, 20) \end{aligned}$$

$$\begin{aligned} \rightarrow F &= P(1+i)^n = 10000(1+0.08)^{20} \\ &= \$46609.57 \end{aligned}$$

2nd Factors (P/A or A/P) (Find/given)
Present worth and Capital Recovery

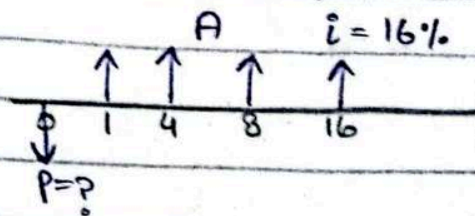
$$A = P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] \text{ and}$$

$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$



Example 2.3

$A = \$600$, $n = 9$, $i = 16\%$, $P = ?$



$$P = \$600(P/A, 0.16, 9)$$

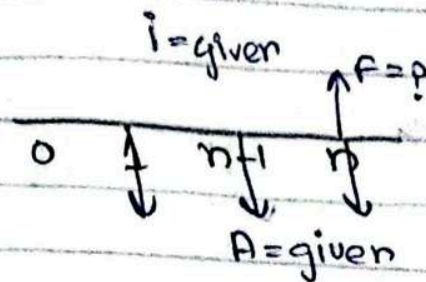
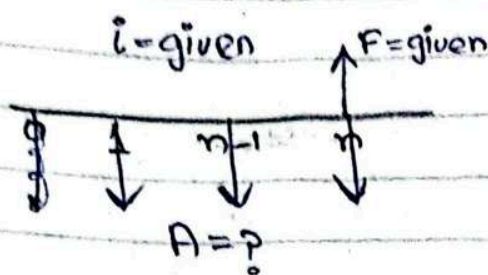
$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] = (600) \left[\frac{(1+0.16)^9 - 1}{(0.16)(1+0.16)^9} \right]$$

$$P = \$2763.92$$

3rd Factor: (F/A or A/F)

$$A = F \left[\frac{i}{(1+i)^n - 1} \right] \text{ and}$$

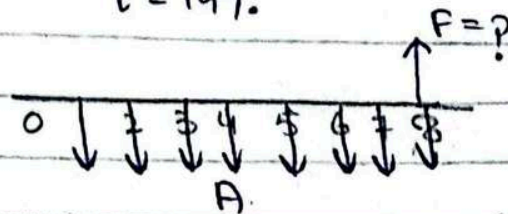
$$F = A \left[\frac{(1+i)^n - 1}{i} \right]$$



Example 2.5

$A = \$1,000,000$, $n = 8$, $i = 14\%$.
starting 1 year

$i = 14\%$



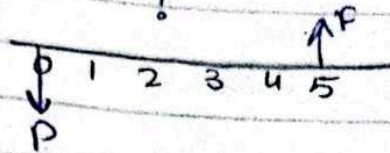
$$F = 1,000,000 (F/A, 0.14, 8)$$

$$F = A \left[\frac{(1+i)^n - 1}{i} \right] = [1,000,000] \left[\frac{(1+0.14)^8 - 1}{0.14} \right]$$

$$F = \$13.2 \text{ million}$$

Example:

$$P = 30,000, n = 5, F = \$50,000, i = ?$$



$$F = P(F/P, i, n)$$

$$50,000 = 30,000(F/P, i, 5)$$

$$F = P(1+i)^n$$

$$\frac{F}{P} = (1+i)^n$$

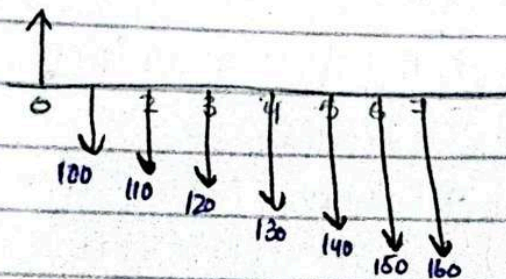
$$\sqrt[n]{\frac{F}{P}} = 1+i$$

$$i = \sqrt[n]{\frac{F}{P}} - 1 = \sqrt[5]{\frac{50,000}{30,000}} - 1$$

$$i = 10.76\%$$

⑤

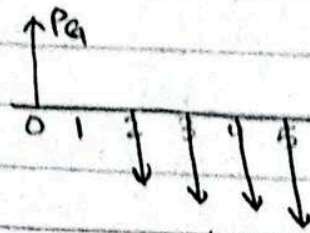
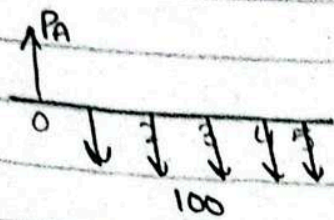
Arithmetic Gradient (A/G, P/G)



$$P_T = (100)(P/F, 10\%, 1) + 110(P/F, 10\%, 2) + 120(P/F, 10\%, 3) \dots$$

$$\rightarrow P_G = \frac{G}{i} \left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n} \right]$$

$$\rightarrow A_G = G \left[\frac{1}{i} - \frac{n}{(1+i)^n - 1} \right] \therefore \text{gradient} = 10$$

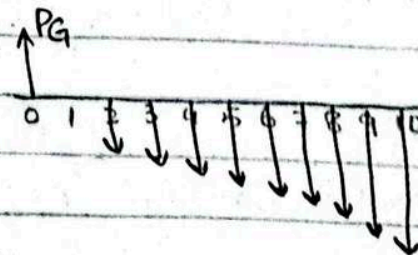
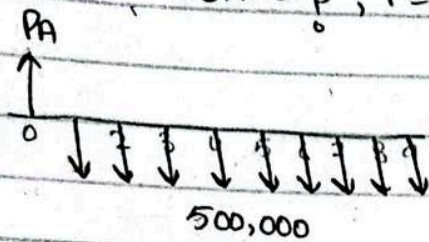


$$P_T = P_A + P_G$$

$$P_A = 100 (P/A, 10\%, 5)$$

$$P_G = 10 (P/G, 10\%, 5)$$

Example: Deposit \$500,000 next year, \$100,000 increase for 9 years - Calculate present worth = P , $i = 5\%$.



$$P_T = P_A + P_G$$

$$= 500,000 [P/A, 5\%, 10] + 100,000 (P/G, 5\%, 10)$$

$$= 500,000 \left[\frac{(1+0.05)^{10} - 1}{(0.05)(1+0.05)^{10}} \right] + \frac{100,000}{5} \left[\frac{(1+0.05)^{10} - 1}{(0.05)(1+0.05)^{10}} - \frac{10}{(1+0.05)^{10}} \right]$$

$$= \$7026.05$$

$$\rightarrow AG = G \left[\frac{1}{i} - \frac{n}{(1+i)^n - 1} \right]$$

$$AT = AA + AG$$

$\therefore$ It is same annual to annual

(6)

$\therefore$ Series gradually increases by fixed value is Arithmetic series.

$\rightarrow$ It increases linearly.

Geometric series:

Increasing by a fixed percentage.

$\rightarrow$ It increases exponentially.

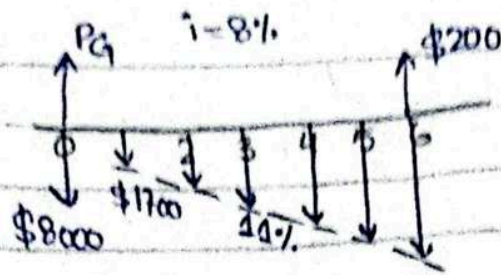
$$\Rightarrow Pg = A(P/A, g, i, n)$$

$$(P/A, g, i, n) = \begin{cases} \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} & i \neq g \\ \frac{n}{1+i} & i = g \end{cases}$$

i and g will be decimals.

Example 2.11

cost = \$8000 ; $n = 6$ years ; \$200 salvage value
\$1700 in 1st year ; $g = 11\%$; $i = 8\%$.



$$= \frac{1 - \left(\frac{1+0.11}{1+0.8} \right)^6}{0.8 - 0.11}$$

$$P = -8000 - 1700(P/A, 11\%, 8\%, 6) + 200(P/F, 8\%, 6)$$

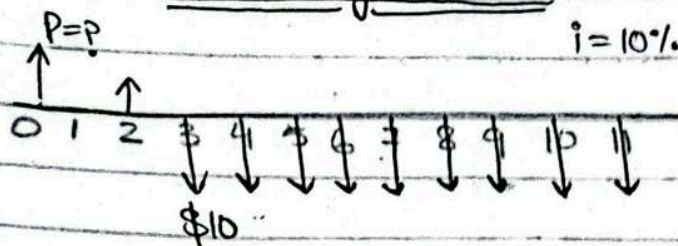
$$= -8000 - 1700 \left[\frac{1 - \left(\frac{1+0.11}{1+0.8} \right)^6}{0.8 - 0.11} \right] + \frac{200}{(1.08)^6}$$

$$= \$+17,999$$



Chapter no 3

Combining Factors



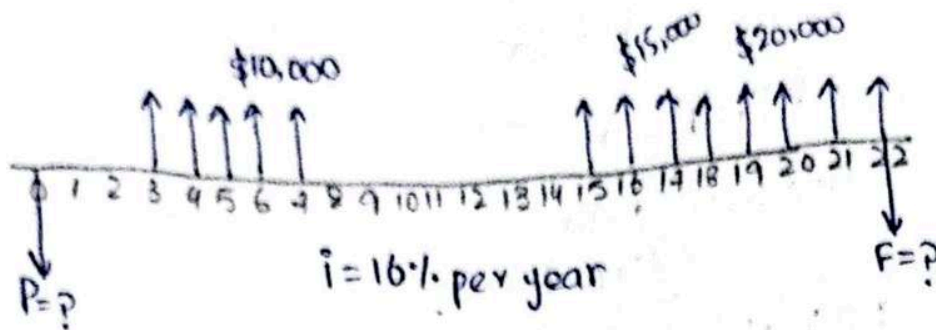
$$P = 10(P/A, 10\%, 8)(P/F, 10\%, 2)$$

$$P = 10 \left[\frac{(1-0.1)^8}{(0.1)(1.01)^8} \right] \left[\frac{1}{(1.01)^2} \right]$$

$$P = \$98.02$$

Name: Sara Said
 Roll no: 221-7734
 BS-EE-6C
 Engineering Economics

Example # 3.4



① Total Present worth P_T in year 0:

$$\begin{aligned}
 P_T &= 20000(P/A, 16\%, 20)(P/F, 16\%, 2) + 10000(P/F, 16\%, 6) \\
 &\quad + 15000(P/F, 16\%, 16) \\
 &= 20,000 \left[\frac{(1+0.16)^{20}-1}{(0.16)(1+0.16)^{20}} \right] \left[\frac{1}{(1+0.16)^2} \right] + 10000 \left[\frac{1}{(1+0.16)^6} \right] + 15000 \left[\frac{1}{(1+0.16)^{16}} \right] \\
 &= 88,121.89 + 4104.42 + 1395.60 \\
 &= \boxed{\$93,621.92}
 \end{aligned}$$

② Future worth F in 22 years:

$$\begin{aligned}
 F &= 20,000(F/A, 16\%, 20) + 10,000(F/P, 16\%, 16) + 15,000(F/P, 16\%, 6) \\
 &= 20000 \left[\frac{(1+0.16)^{20}-1}{0.16} \right] + 10000[1+0.16]^{16} + 15000(1+0.16)^6 \\
 &= 2307594.93 + 107480.042 + 36545.94 \\
 &= \boxed{\$2,451,621}
 \end{aligned}$$

③ Annual series over all 22 years:

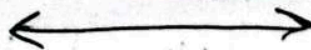
$$\begin{aligned}
 A &= F_T(A/F, 16\%, 22) \\
 &= 2,451,621 \left(\frac{0.16}{(1+0.16)^{22}-1} \right) \\
 &= \boxed{\$15,574.2}
 \end{aligned}$$

② Annual series over the first 10 years:

$$\begin{aligned} A_{1-10} &= P_t (A/P, 16\%, 10) \\ &= 93,621 \left[\frac{0.16(1+0.16)^{10}}{(1+0.16)^{10}-1} \right] \\ &= \$19,163 \end{aligned}$$

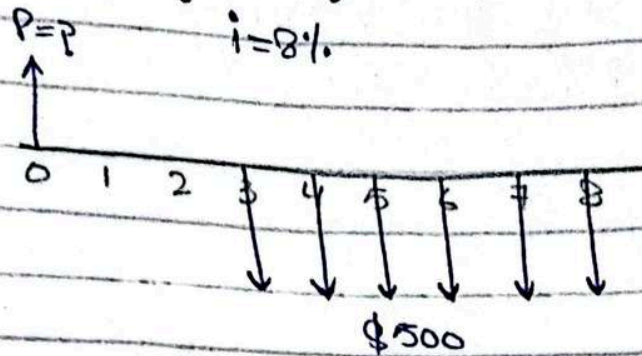
③ Annual series over the last 12 years:

$$\begin{aligned} A_{11-22} &= F(A/F, 16\%, 12) \\ &= 2,451,621 \left[\frac{0.16}{(1+0.16)^{12}-1} \right] \\ &= \$79,468,64 \end{aligned}$$



Example # 3.1

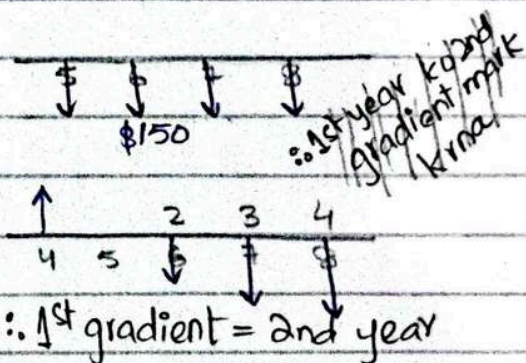
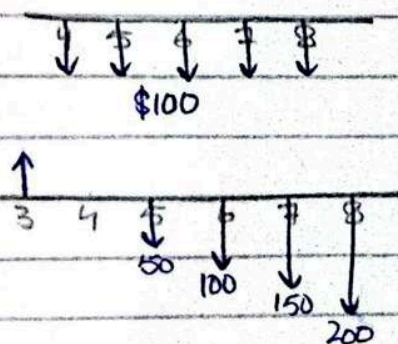
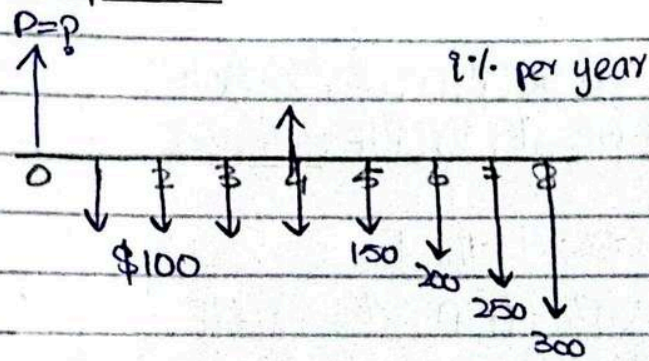
$A = \$500$, $n = 6$ years, starting from 3rd year,
 $P = ?$ at 0th year, for $i = 8\%$.



$$\begin{aligned}
 P &= 5000 + 500(P/A, 8\%, 6)(P/F, 0.08, 2) \\
 &= 5000 + 500(4.6229)(0.8573) \\
 &= \$6981.60
 \end{aligned}$$

(7)

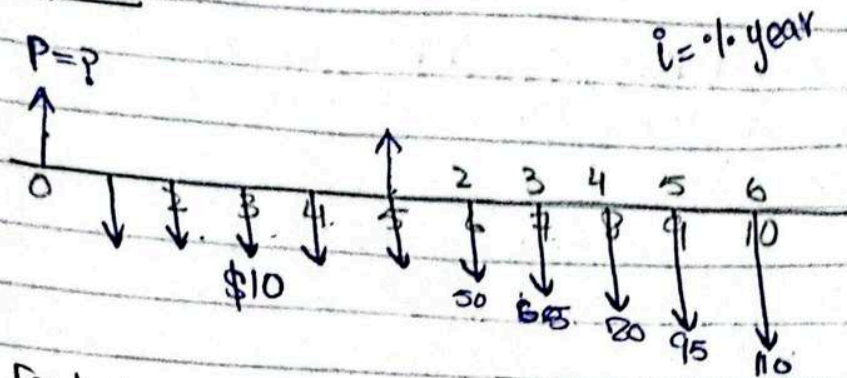
Arithmetic Gradient:



$$P_T = 100(P/A, i\%, 4) + 50(P/G, i\%, 4)(P/F, i\%, 4) \\ + 150(P/A, i\%, 4)(P/F, i\%, 4)$$

$$P_T = 100(P/A, i\%, 3) + 100(P/A, i\%, 5)(P/F, i\%, 3) \\ + 50(P/G, i\%, 5)(P/F, i\%, 3)$$

Example:



∴ First break the diagram into 2 cashflows

$$P_T = 10(P/A, i\%, 5) + 50(P/A, i\%, 5)(P/F, i\%, 5) \\ + 15(P/G, i\%, 5)(P/F, i\%, 5)$$

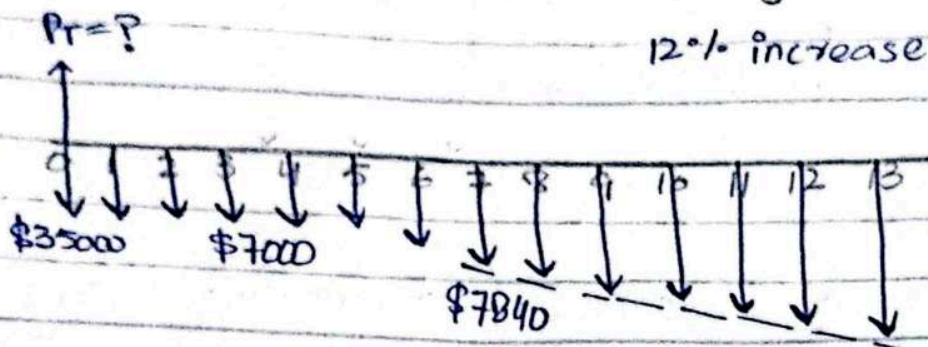
Alternative Method

$$P_T = 10(P/A, i\%, 10) + 40(P/A, i\%, 5)(P/F, i\%, 5) \\ + 15(P/G, i\%, 5)(P/F, i\%, 5)$$

Geometric Gradient

Example # 3.7

\$7000 per year starting 1 year $\rightarrow$ 5 year
Annual price increases 12% per year after \$35000
at 0, $i = 15\%$, $n = 5$, 8 year



1st Method: 7000 se increase kar de

2nd Method: 7840 se increase kar de

$$Pr = 35000 + 7000(P/A, 15\%, 5) + 7840[(P/A, 12\%, 15\%, 8) \times (P/F, 15\%, 5)]$$

$$= 35000 + 7000(2.8550) + \left[7000 \frac{1 - (1.12/1.15)^9}{0.15 - 0.12} \right] (0.5718)$$

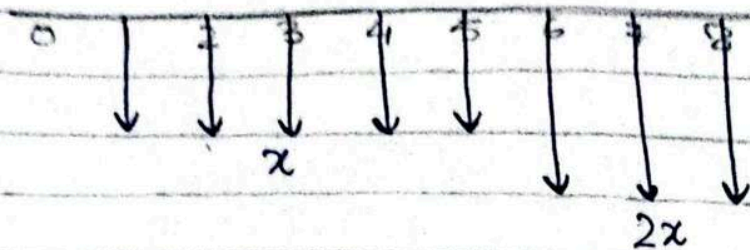
$$= 35000 + 19985 + 28247$$

$$= \$83,232$$

Exercise # 3.27

For the cashflow diagram, determine the value of x that will make the future worth in 8 years equal to \$-70000.

$i = 10\%$ per year



$$70000(P/F, i\%, 8) = x(P/A, i\%, 5) + 2x(P/A, i\%, 3)(P/F, i\%, 5)$$
$$70000 \left(\frac{1}{(1+0.10)^8} \right) = x \left(\left[\frac{(1+0.10)^5 - 1}{(0.10)(1.10)^5} \right] \right) + 2x \left[\frac{(1.10)^3 + 1}{(0.10)(1.10)^3} \right] \left(\frac{1}{(1.10)^5} \right)$$

$$70000(0.46) = x(3.79) + 2x[(2.48)(0.62)]$$

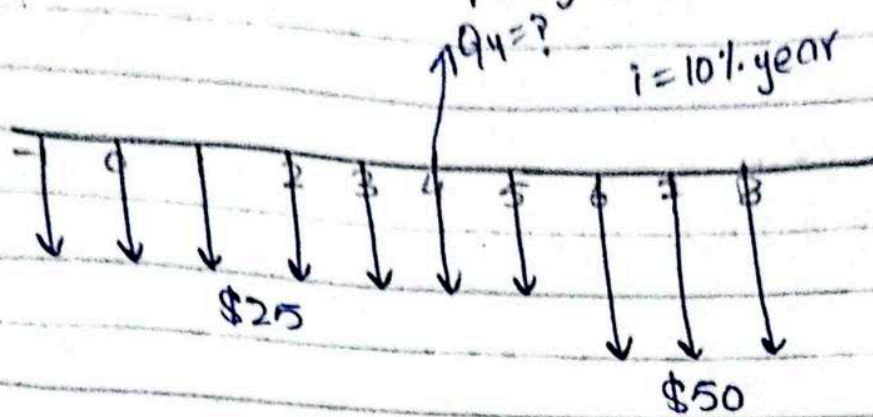
$$3220 = 3.79x + 3.07x$$

$$3220 = 6.8652x \Rightarrow x = \frac{3220}{6.8652} \Rightarrow \boxed{x = \$469}$$

Exercise # 3.31

Use cashflow diagram to determine the single amount of money Q_4 in year 4 that

is equivalent to all of the cash flows -
 use $i = 10\%$ per year.

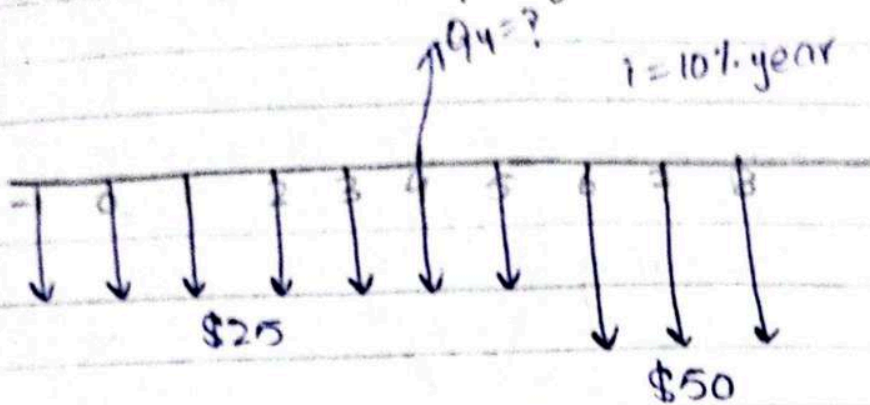


$$= 25(P/A, i\%, 7)(P/F, i\%, 1) + 50(P/A, i\%, 3)(P/F, i\%, 1)$$

$$= 25 \left[\frac{(1.1)^7 - 1}{(0.1)(1.1)^7} \right] \left[\frac{1}{(1.1)^1} \right] + 50 \left[\frac{(1.1)^3 - 1}{(0.1)(1.1)^3} \right] \left[\frac{1}{(1.1)^1} \right]$$

$$= 25(4.42) + 50(2.26) = \$223.5$$

is equivalent to all of the cash flows.
Use $i = 10\%$ per year.



$$= 25(P/A, i\%, 7)(P/F, i\%, 1) + 50(P/A, i\%, 3)(P/F, i\%, 1)$$

$$= 25 \left[\frac{(1.1)^7 - 1}{(0.1)(1.1)^7} \right] \left[\frac{1}{(1.1)^1} \right] + 50 \left[\frac{(1.1)^3 - 1}{(0.1)(1.1)^3} \right] \left[\frac{1}{(1.1)^1} \right]$$

$$= 25(4.42) + 50(2.26) = \$223.5$$

Chapter no 4

Nominal & Effective Interest Rate

Nominal Interest Rate:

This is basic interest rate that doesn't consider compounding - It is simply rate on a loan or investment.

For example: if a bank offers a 12% nominal annual interest rate, that means the bank is

* r = interest rate per time period $\times$ number of periods

Saying 12% per year, but it doesn't tell you how often the interest rate is applied.

Effective Interest Rate:

It is the real interest rate you actually earn or pay after compounding.

$$i = EIR = \left(1 + \frac{r}{m}\right)^m - 1$$

i = effective rate per specified time period
 r = nominal rate per same time period
 m = no. of compounding periods per year

Question no 1

Bit 1: 9%, per year, compounded quarterly

Bit 2: 3%, per year, compounded quarterly

Bit 3: 8.8%, per year, compounded monthly

Effective rate for 6 months

Bit 1:

4.5% / 6 months

$\therefore$ 9 ka quarter 2.25
 and 225 ka 2
 4.5 (6 months)

$$i \text{ for 6 months} = \left(1 + \frac{r}{n}\right)^n - 1$$

$$= \left(1 + \frac{0.045}{2}\right)^2 - 1$$

$$= 4.55\% / 6 \text{ months}$$

Bit 2:

Now 3% quarterly will be 0.75
 and for 6 months, $2 \times 0.75 = 1.5$

$$i = \left(1 + \frac{0.015}{2}\right)^2 - 1$$

$$i = 1.50\% / 6 \text{ months}$$

Bit 3:

8.8% monthly will be 0.73 and for 6 months, $6 \times 0.73 = 4.4$

$$So, i = \left(1 + \frac{0.044}{6}\right)^6 - 1$$

$$i = 4.48\% / 6 \text{ months}$$

$\rightarrow$ Effective rate for a year

Bit 1:

$$\left(1 + \frac{0.09}{4}\right)^4 - 1 = 9.3\% / \text{year}$$

Bit 2:

$$\left(1 + \frac{0.12}{4}\right)^4 - 1 = 12.5\% / \text{year}$$

Bit 3:

$$\left(1 + \frac{0.088}{12}\right)^{12} - 1 = 9.1\% / \text{year}$$

Equivalence: Payment Period & Compounding Period

Equivalence	Payment Period	Compounding
Different interest rates & payment structures can result in the same overall value of time.	This is how often payments are made.	This is how often interest is added to the principal.

- If payment period & compounding period are same, calculations are straightforward.
- If they're different, we need to adjust the interest rate so that the effective rate over payment period is correctly calculated.

* Single amount with $PP \geq CP$

Method 1:

$$P = F (PIF, \text{effective } i/CP, \text{total no. of provided } m)$$

Question:

$$F = ? ; P = \$1000 ; n = 2 \text{ years}$$

$i = 15\%$, compounded monthly

$$F = 1000 (FIP, 1.25\% / \text{monthly}, 24)$$

$$F = 1000 (1 + 1.25\%)^{24}$$

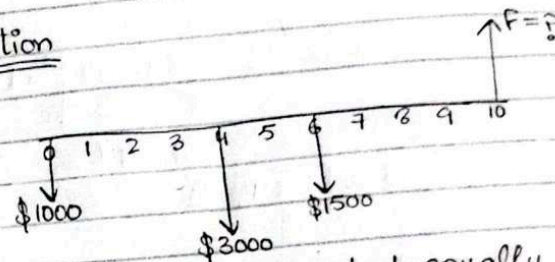
$$F = 1347.35$$

Method 2: $i = \left(1 + \frac{0.15}{12}\right)^{12} - 1$

$$i = 16.07\% / \text{year}$$

$$F = 1000 (FIP, 16.07\%, 2) = 1000 (1 + 0.01607)^2 = \$1347.22$$

* Question



$i = 12\%$ year, compounded annually

Method 1:

$$F = 1000 (FIP, 6\%, 20) + 3000 (FIP, 6\%, 12) + 1500 (FIP, 6\%, 8)$$

$$F = 1000 (1 + 0.06)^{20} + 3000 (1 + 0.06)^{12} + 1500 (1 + 0.06)^8$$

$$F = \$11634.497$$

Method 2:

$$i = \left(1 + \frac{0.12}{2}\right)^2 - 1 = 12.36\% / \text{year}$$

$$F = 1000 (FIP, 12.36\%, 10) + 3000 (FIP, 12.36\%, 6) + 1500 (FIP, 12.36\%, 4)$$

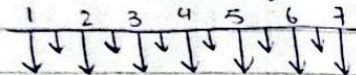
$$F = 1000 (1.1236)^{10} + 3000 (1.1236)^6 + 1500 (1.1236)^4$$

$$F = \$11634.497$$

→ $PP > CP$ for series

Find the effective i per PP and let n the total no. of payment period

$n=7$; \$500 every 6 months ; 8%



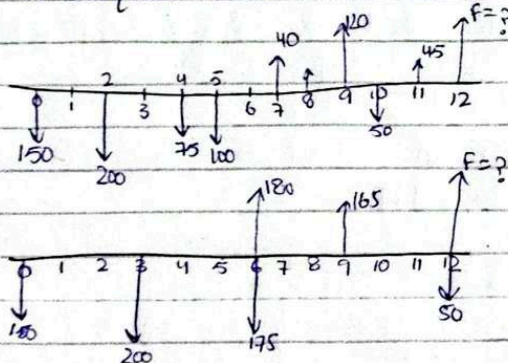
$$\text{effective rate per 6 months} = \left(1 + \frac{0.08}{2}\right)^2 - 1$$

$$= 4.04\%$$

$$\text{So, } F = 500 (F/A, 4.04\%, 14)$$

→ $PP < CP$ (single amount & series)

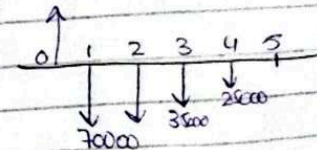
No interperiod compounding
All the outflows will be moved at the end of CP



$$F = 1000 [150(F/P, 3\%, 4) - 200(F/P, 3\%, 3) - 175(F/P, 3\%, 2) + 180(F/P, 3\%, 2) + 165(F/P, 3\%, 1) - 50]$$

Interest Rate vary with time:

Year	1	2	3	4
Net Profit	70,000	70,000	35,000	25,000
Annual Rate	7%	7%	9%	10%



$$P = F_1(P/F, i_1, 1) + F_2(P/F, i_1, 1)(P/F, i_2, 1) + F_3(P/F, i_1, 1)(P/F, i_2, 1)(P/F, i_3, 1)$$

$$= 70000(P/F, 7\%, 1) + 70000(P/F, 7\%, 1)(P/F, 7\%, 1)$$

$$+ 35000(P/F, 9\%, 1)(P/F, 7\%, 1)(P/F, 7\%, 1)$$

$$+ 25000(P/F, 10\%, 1)(P/F, 9\%, 1)(P/F, 7\%, 1)$$

Chapter no 5

Present Worth Analysis

- ↳ Mutually exclusive project
- ↳ Independent project

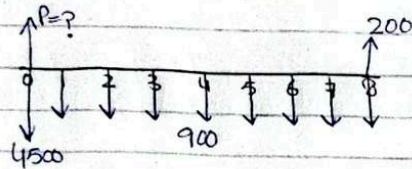
→ Revenue based

→ Cost based

Multiple Mutually Exclusive

MARR 10% year	Electric Power	Gas Power	Solar Power
First Cost, \$	4500	3500	6000
Annual Operating	900	700	50
Salvage value	200	350	100
Life (years)	8	8	8

Electric Power:



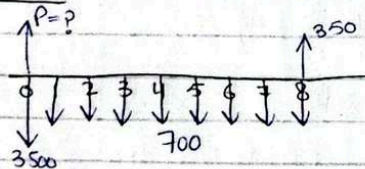
$$P_T = -4500 + 900(P/A, 10\%, 8) + 200(P/F, 10\%, 8)$$

$$P_T = -4500 - 900(P/A, 10\%, 8) + 200(P/F, 10\%, 8)$$

$$P_T = -4500 - 900 \left[\frac{(1.10)^8 - 1}{(0.1)(1.1)^8} \right] + 200 \left[\frac{(1.10)^8 - 1}{(0.1)(1.1)^8} \right]$$

$$P_T = \$-9200$$

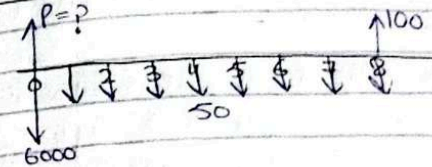
Gas Power:



$$P_T = -3500 - 700(P/A, 10\%, 8) + 350(P/F, 10\%, 8)$$

$$P_T = \$-7071$$

Solar Power:



$$P_T = -6000 - 50(P/A, 10\%, 8) + 100(P/F, 10\%, 8)$$

$$P_T = \$-6220$$

We prefer solar power for 8 years

But what will happen if we increase or decrease life span?

if lifespan is 4 years

then

$$\text{Electric: } P_T = \$-7218$$

$$\text{Gas: } P_T = \$-5479$$

$$\text{Solar: } P_T = \$-6090$$

} Multiple Mutually exclusive

So when we decrease the Lifespan, Gas Power becomes more suitable. So this is how Lifespan can change the results.

What will happen if the Lifespan is different?

For the case of Multiple Mutually exclusive with different lifespans there are two methods:

① Take LCM of their lives
Phir jo answer aye usko total life Le Lo

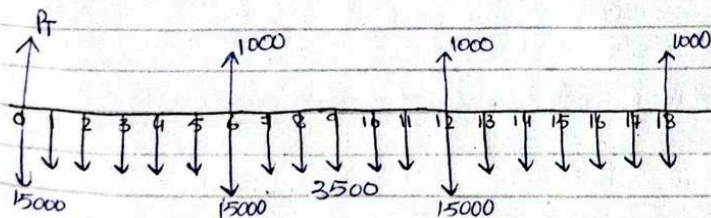
② Study period method (unreasonable high life)
Kisi b 1 time period k liye sab ko solve kar do jo in se bahar ho.

Example #5.3

	Vendor A	Vendor B
First Cost, \$	-15,000	-18,000
Annual M&O	-3500	-3100
Salvage value	1000	2000
Life, years	6	9

Taking LCM = 18
So, now we will solve both for 18 years

Vendor A:



$$P_W = -15000 - 3500(P/A, 15\%, 18) - 15000(P/F, 15\%, 6) \\ - 15000(P/F, 15\%, 12) + 1000(P/F, 15\%, 6) \\ + 1000(P/F, 15\%, 12) + 1000(P/F, 15\%, 18)$$

But I will prefer first method :-

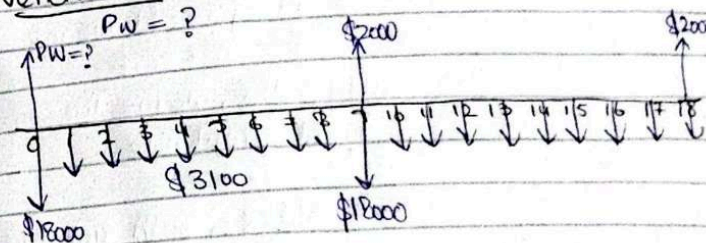
OR

We have same value at point such as (15000, 1000) add / subtract them

$$P_W = -15000 - 3500(P/A, 15\%, 18) - 14000(P/F, 15\%, 6) \\ - 14000(P/F, 15\%, 12) + 1000(P/F, 15\%, 18)$$

$$P_W = \$ -267.36$$

Vendor B:



$$P_W = -18000 - 3100(P/A, 15\%, 18) - 18000(P/F, 15\%, 9) \\ - 18000(P/F, 15\%, 18) + 2000(P/F, 15\%, 9) + 2000(P/F, 15\%, 18)$$

$$P_W = -18000 - 3100(P/A, 15\%, 18) - 18000(P/F, 15\%, 9) \\ + 2000(P/F, 15\%, 9) + 2000(P/F, 15\%, 18)$$

$$P_W = \$ 27.397$$

FUTURE WORTH ANALYSIS

Future worth analysis helps in decision making by evaluating the long-term value of different alternatives, ensuring that the best investment choice is made.

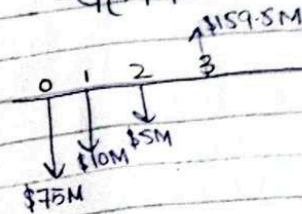
- ∴ the selection of future worth analysis is same as present worth analysis. It means
 - $FW = 0$
this project earn exactly MARR, means the project is suitable but not profitable
 - $FW > 0$
the project exceeds MARR, both suitable and profitable
 - If there are two or more alternatives, select the one with highest FW.

Example # 5.5

- Initial Investment = \$75 million
- Net loss at year 1 = \$10 million ($A_1 = -10$)
- Cash flow increases with $AG = \$5$ million from 2 year
- MARR = 25%
- OFFER PRICE (year 3) = \$159.5 million

a) Will selling now for \$159.5 million meet the 25% MARR?

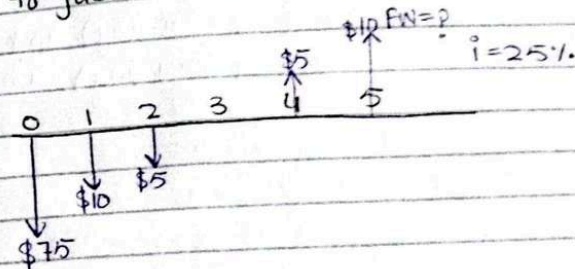
$$FW = -P(FIP, 25\%, 3) - A_1(FIA, 25\%, 2) - G(FIG, 25\%, 1) + \text{Selling price}$$



$$FW = \$8.86 \text{ Million}$$

Since FW is -ve, the required 25% is not met at selling price.

b) What selling price is needed at year 5 to just meet the MARR?



$$FW = -P(FIP, 25\%, 5) - A_1(FIA, 25\%, 5) + G(FIG, 25\%, 5)(FIA, 25\%, 5)$$

$$FW = -75(3.052) - 10(7.75) + 5(11.9)(7.75)$$

$$FW = \$+246.81 \text{ Million}$$

Capitalize Cost Analysis (CC)

→ Project has very long life

$$\rightarrow P = A \left[\frac{1 - (1/(1+i)^n)}{i} \right]$$

$$CC = \frac{A}{i} \Rightarrow A = CC \times i$$

Procedure to determine CC for an infinite sequence of cash flows:

1- Draw a cashflow diagram

- Show one-time costs (non-recurring) and at least two cycles of repeating (recurring) cashflows.

2- Find PW of Nonrecurring Amounts

- Calculate PW of one-time costs
- This is your 1st part of CC.

3- Find Annual worth A of recurring costs

- Calculate A through one full cycle of repeating cost (e.g. every 5 years)
- If there are other annual costs, add them too → This gives you total AW

4- Divide AW by Interest Rate (i)

- Use $CC = \frac{AW}{i}$

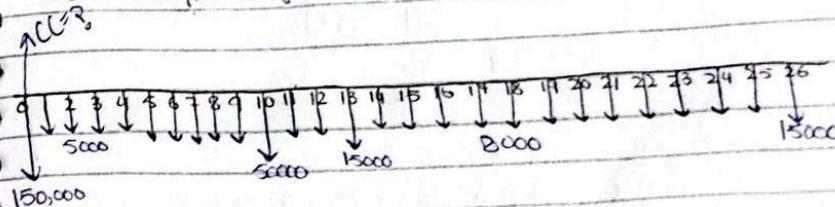
5- Add CC values

- Add both value from step 2 and step 4.

Example # 5.6

\$150,000 at year 0
\$50,000 additional cost at year 10
\$5000/year for 4 years, then \$8000/year
\$15000 upgrade every 13 years (start from 13th year)

* Draw cashflow diagram



* PW of nonrecurring value (50,000)

$$\begin{aligned} CC_1 - PW &= -150,000 - 50,000(P/F, 5\%, 10) \\ &= -150,000 - 50,000 \left[\frac{1}{(1.05)^{10}} \right] \\ &= \$-180,695 \end{aligned}$$

* Calculate A for recurring cost (15000)

$$A = -15000 (A/F, 5\%, 13)$$

$$= -15000 \left[\frac{0.05}{(1.05)^{13} - 1} \right] = -846.83$$

$$CC_2 = \frac{-847}{0.05} = -16,936$$

* Any additional costs (5000 & 8000)

$$CC_3 = \frac{-5000}{0.05} = -100,000$$

$$\left(\frac{5000+3000}{=8000} \right) CC_4 = \frac{-3000}{0.05} = -60,000$$

$$CC_4 = -60000 (P/F, 5\%, 4)$$

$$= -60000 \left[\frac{1}{(1.05)^4} \right]$$

$$= -49,362$$

So,

Now $CC_T = CC_1 + CC_2 + CC_3 + CC_4$ (Final Step)

$$CC_T = -180,695 + (-16,936) + (-100,000) + (-49,362)$$

$$CC_T = \$-346,993$$

b) AW value?

$$AW = P \cdot i = CC_T \times i$$

$$= (-346,993) \times (0.05)$$

$$= \$-17,349.65$$



Chapter # 6

Annual Worth Analysis

- When comparing projects that repeat over time, the AW method is more suitable
- You only need to calculate AW for one life cycle, and the same value applies to all future cycles
- Once you find AW for one cycle, you can use it to compare alternatives directly, no repeating, no adjusting for different life spans!

Example # 6.1

First Cost = \$15,000

Annual M&O cost = \$3500/year

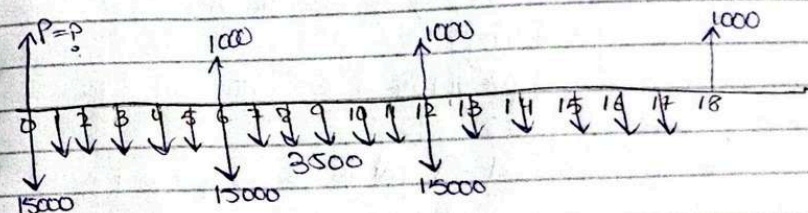
Salvage value at year 6 = \$1000

Interest rate = 15%

Life = 6 years

3 cycles for 18 years

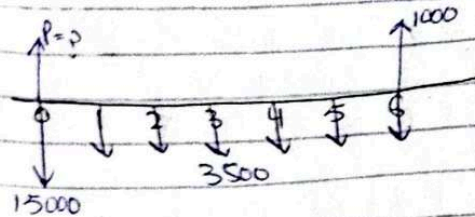
PW over 18 years is \$-45,036



So AW helps us to find, that instead for all 18 years, we can do calculation for just 1 cycle

(6 years), it will remain same forever.

So,



$$A = -15000 - 3500$$

$$A = -15000(A/P, 15\%, 6) - 3500 + 1000(A/F, 15\%, 6)$$

$$A = -15000 \left[\frac{(0.15)(1.15)^6}{(1.15)^6 - 1} \right] - 3500 + 1000 \left[\frac{0.15}{(1.15)^6 - 1} \right]$$

$$A = \$-7,349/\text{year}$$



Chapter # 9

Benefit Cost Analysis & Public Sector Economics

Benefit-cost Analysis is a method used to help decide if a project is worth doing by comparing:

- the benefit it provides
- the cost it takes

It often used in Public sector projects, but also in business

B/C ratio	Means	Decision
$B/C > 1$	Benefits > costs	Accept
$B/C = 1$	Benefits = costs	Marginal
$B/C < 1$	Benefits < costs	Reject

Benefit-Cost Ratio B/C:

$$B/C = \frac{PW \text{ of Benefit}}{PW \text{ of Cost}} = \frac{AW \text{ of B}}{AW \text{ of C}} = \frac{FW \text{ of B}}{FW \text{ of C}}$$

$$\rightarrow \text{Conventional B/C} = \frac{\text{benefit} - \text{disbenefit}}{\text{Cost}}$$

$$\rightarrow \text{Modified B/C} = \frac{\text{benefit} - \text{disbenefit} - \text{M\&O cost}}{\text{Initial investment}}$$

$$\rightarrow \text{Benefit cost difference} = B - C$$

Example # 9.2

- Award cost: \$20 million at end of year 1, decreasing by \$5M/year for 3 more years
- Annual costs: \$2 M/year for 10 years
- Benefits: \$8 M per year
- Disbenefits: \$0.1M to \$0.6M/year
- $i = 6\%$
- Time = 10 years

Step 1: Convert all domains in one
It is a decreasing arithmetic gradient
for 4 years
 $AW = 20 - 5(A/G, 6\%, 4)$
 $= \$12,864/\text{year}$

Step 2: B/C calculations
→ Conv. B/C:
$$= \frac{B - DB}{AW + \text{Annual}} = \frac{8 - 0.6}{12.864 + 2} = 0.50$$

 $0.50 < 1$ (rejected)

→ Mod. B/C:
$$= \frac{B - DB - \text{Annual}}{AW} = \frac{8 - 0.6 - 2}{12.864}$$

 $= 0.42$
 $0.42 < 1$ (rejected)

Normally hum conventional method use
kreenge ☺

Incremental Cost ΔC :

Incremental cost is the additional cost
you have to pay when choosing one option
over another

→ Simply, if I go with Option B instead of
Option A, how much will it cost me?
And is it worth it?

→ Incremental Cost $\Delta C = \text{Cost of Higher} - \text{Cost of Lower}$

→ Incremental Benefit $\Delta B = \text{Benefit of higher} - \text{Benefit of Lower}$

→ Then we calculate B/C ratio:
$$B/C = \frac{\Delta B}{\Delta C}$$

Selection:
• if ratio ≥ 1 (Select higher total cost)
• if ratio < 1 (Select Lower total cost)

Example # 9.4

	Design A	Design B
Construction Cost	10,000,000	15,000,000
Building maintenance	35,000	55,000
Patient Usage Copay	450,000	200,000
AW cost	685,500	1,030,750
Disbenefits	0	100,000
Net Benefits	-	150,000
B/C ratio	-	0.43

(C)

1- First we will calculate AW for both designs
 $AW_A = 10,000,000(A/P, 5\%, 30) + 35,000$
 $= \$685,500$

$$AWB = 15,000,000(A/P, 5\%, 30) + 55,000$$

$$= \$1,030,750$$

2- Find Incremental Cost ΔC

$$\Delta C = AWB - AWA$$

$$= 1,030,750 - 685,500 = 345,250$$

3- Find Incremental Benefit ΔB

$$\Delta B = 450,000 - 200,000 = 250,000/\text{year}$$

4- B/C ratio

$$= \frac{250,000}{345,250} = 0.72$$

As $B/C < 1$ (Design A will be suitable)

b) When disbenefits are considered

Disbenefits of Design B = \$100,000

But Design B has extra benefits = \$250,000

$$\text{Net Benefit} = 250,000 - 100,000$$

$$= 150,000$$

$$B/C = \frac{150,000}{345,250} = 0.43$$

$B/C < 1 \rightarrow$ still design A is better

$\therefore$ question tabular form mein solve krna hai

How to solve question?

- 1- Bring data in same domain
- 2- Total Cost
- 3- Net Benefit
- 4- High Cost - Low Cost
- 5- High Benefit - Low Benefit
- 6- Check ratio

$\rightarrow$ Full Multiple cases

Challenger $\rightarrow$ High cost
Defender $\rightarrow$ Low Cost
then we compare next values

Example # 9.6

	City 1	City 2	City 3	City 4
First Cost \$	38.5	40.1	45.9	60.3
Entrance fee cost	500,000	450,000	425,000	250,000
Annual revenue	7.0	62	10.0	10.4
Initial cash incentive	250,000	350,000	500,000	800,000
Property tax reduce	25,000	35,000	50,000	80,000
Sales tax sharing	310,000	320,000	320,000	340,000

	City 1	City 2	City 3	City 4
First cost x A/P	6.42	6.69	7.66	10.07
AW of 1st Cost	41750	58450	83500	133600
Initial cash incentive	6.948	7.166	8.112	10.348
AW of total cost	7.377	6.614	10.454	10.954
AW of total Bene	1.06	0.92	1.28	1.05
Overall B/C (total Bene / total cost)				

Alternative compared	City 1	City 2	City 3	City 4
	1-to-ON	B/C < 1	3 to 1	4 to 3
ΔC	6.948		1.164	2.336
ΔB	7.377		3.077	0.50
$\Delta B/C$	1.06		2.64	0.22
Incremental justified	Yes	Eliminated	Yes	No
City Selected	1		3	3

$$i = 7\% ; n = 8 \text{ years} ; A/P = \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$A/P = \frac{(0.07)(1.07)^8}{(1.07)^8 - 1} = 0.167$$

- AW of initial cost incentive = $0.167 \times \text{Initial cost incentive}$
 → AW of total cost = first cost $(A/P, 7\%, 8) + \text{entrance fee}$

$$= 38.5(A/P, 7\%, 8) + 0.5 = \$6.948 \text{ M}$$

→ AW of total benefits = revenue + initial incentive (0.167)
 + property tax + sales tax sharing
 $= 7.0 + 0.25(0.167) + 0.025 + 0.31 = \7.377 M

- ΔC = higher cost - lower cost
 → ΔB = higher benefit - lower benefit
 → $\Delta B/C = \Delta B / \Delta C$

Problem 9.31

	East location	West location
Interest rate 6%		
Initial Cost	11×10^6	27×10^6
Annual Mgt	100,000	90,000
Benefits	990,000	2,400,000
Disbenefits	120,000	100,000
Life	∞	∞
Net Benefits (B-D)	$990,000 - 120,000$ $= 870,000$	$2,400,000 - 100,000$ $= 2,300,000$
Annual Cost (C)	$11,000,000(0.06) + 100,000$ $= 760,000$	$27,000,000(0.06) + 90,000$ $= 1,710,000$
Benefit Cost B/C	$\frac{870,000}{760,000} = 1.14$	$\frac{2,300,000}{1,710,000} = 1.35$

Since both B/C are greater than 1, we now perform Incremental B/C Analysis West - East

$$\text{Incremental Benefits} = 2,300,000 - 870,000$$

$$\Delta(B-D) = 1,430,000$$

$$\text{Incremental Cost} = 1,710,000 - 760,000$$

$$\Delta C = 950,000$$

$$\text{Incremental } \Delta B/C = \frac{1,430,000}{950,000} = 1.51$$

West wing is suitable

Chapter no 17

After Tax Economic Analysis

* Taxable Income = Gross Income - Operating Expenses
- Depreciation

$$TI = GI - OE - D$$

First-Next	Tax	From	Range	TI Range	Tax Inversed
150,000	8%	1	150,000	12000	12000
200,000	12%	150,001	350,000	24000	36000
200,000	18%	350,001	550,000	36000	72000
150,000	24%	550,001	700,000	36000	108,000
250,000	30%	700,001	950,000	75000	183,000
300,000	40%	950,001	1,250,000	120,000	303,000
500,000	42%	1,250,001	1,750,000	210,000	513,000
Above	50%				

$$\begin{aligned} 900,000 &= 150,000(0.08) + 200,000(0.12) \\ &\quad + 200,000(0.18) + 150,000(0.24) + 250,000(0.3) \\ &= 168,000 \end{aligned}$$

Chapter no 11

Replacement and Retention Decisions

- Defender → Amount you actually have
- Challenger → Amount that is gonna replace the defender

∴ Challenger is better than defender

Economic Service Life → ESL

↳ ESL means the best time period to use a machine or equipment before it gets too costly to keep. It includes buying cost, maintenance and operation. When these costs get too high, it's better to replace the machine.

Sunk Cost:

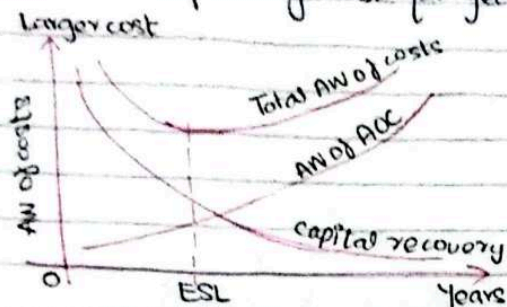
It is the money that has already been spent and cannot be recovered. It should not affect the future decisions because it won't change.

Example: You bought a movie ticket for Rs 500 but feel sick and don't want to go, that Rs 500 is a sunk cost because it cannot be recovered.

Formula:

$$\text{Total } AW_k = -P(A/P, i, k) + S_k(A/F, i, k) - \left[\sum_{j=1}^{j=k} AOC_j(P/F, i, j) \right] \cdot (A/P, i, k)$$

- ① $P \rightarrow$ initial investment / current market value
- ② $S_k \rightarrow$ Salvage value / market value after k years
- ③ $AOC_j \rightarrow$ Annual operating cost for year j ($j=1$ to k)



This graph shows how to find ESL of an asset using AW of cost over time.

Key elements in graph:

1- AW of costs:

Represents the yearly cost of owning and operating the asset, shown as AW.

2- Capital Recovery:

The decreasing curve represents the owning the asset (initial cost - salvage). It decreases over time because the initial cost is spread over more years.

$$= -P(A/P, i, n) + S(A/P, i, n)$$

3- AW of AOC:

Increasing curve, over time as machine gets older.

AOC = Annual operating cost

4- Total AW (U-shaped curve):

Sum of Capital Recovery + AW of AOC

5- ESL:

Lowest point of Total AW is ESL.

Example # 11.2

Year (j)	MV _j , \$	AOC _j , \$
1	10,000	-5,000
2	8,000	-6,500
3	6,000	-8,000
4	2,000	-9,500
5	0	-12,500

current market value = \$20,000

$i = 10\%$

MV_j = Salvage Value ; AOC = Annual operating

Now,

$$AW_{total} = \text{Capital Recovery} + \text{AW of AOC}$$

where

$$\text{Capital Recovery} = -P(A/P, i, j) + S_j(A/P, i, j)$$

Ab sir ne yahn kaha tha k saary factors calculate kar k table bna lo.

Personally mujhe yeh kaam bilkul nahi pasand 😊

Year	$A/P, 10\%, j$	$A/F, 10\%, j$
1	1.1	1
2	0.5762	0.4761
3	0.4021	0.3021
4	0.3155	0.2154
5	0.2638	0.1637

$$\rightarrow \text{For } A/P = \frac{10\% [1 + 10\%]^j}{(1 + 10\%)^j - 1}$$

$$\rightarrow \text{For } A/F = \frac{10\%}{(1 + 10\%)^j - 1}$$

Year 1:

$$\begin{aligned} CR &= -P(A/P, 10\%, j) + S(A/F, 10\%, j) \\ &= -20000(1.1) + 10000(1) \\ &= -12000 \end{aligned}$$

$$AOC = 5000 \text{ (from table, ignoring } \pm \text{ sign)}$$

$$AW_{\text{total}} = -12000 + 5000 = -17000$$

Alternative method:

$$\begin{aligned} \text{Total } AW_1 &= -20000(A/P, 10\%, 1) + 10000(A/F, 10\%, 1) \\ &\quad - [5000(P/F, 10\%, 1)](A/P, 10\%, 1) \\ &= -17000 \end{aligned}$$

ignore -ve sign
a) AOC

Ye formula paper mein given hota ha

har saal k liye ya list
specifically bola ho uske
life find kar lein.

Year 3:

$$\begin{aligned} \text{Total } AW_3 &= -20000(A/P, 10\%, 3) + 6000(A/F, 10\%, 3) \\ &\quad - [5000(P/F, 10\%, 1) + 6500(P/F, 10\%, 2) + \\ &\quad \quad 8000(P/F, 10\%, 3)](A/P, 10\%, 3) \\ &= \$-12,635 \end{aligned}$$

Example

$$1^{\text{st}} \text{ cost} = 70000$$

$$AOC = \$20,000$$

$$n = 5 \text{ years (max)}$$

$$\text{Salvage value} = \$10,000$$

$$i = 10\%$$

$$ESL = ? \quad AW = ?$$

$$\begin{aligned} AW_1 &= -70000(A/P, 10\%, 1) - 20000 + 10000(A/F, i, 1) \\ &= \$-87,000 \end{aligned}$$

$$\begin{aligned} AW_2 &= -70000(A/P, 10\%, 2) - 20000 + 10000(A/F, i, 2) \\ &= \$-55,200 \end{aligned}$$

$$\begin{aligned} AW_3 &= -70000(0.40) - 20000 + 10000(0.302) \\ &= \$-44,980 \end{aligned}$$

$$\begin{aligned} AW_4 &= -70000(0.31) - 20000 + 10000(0.21) \\ &= \$-39,600 \end{aligned}$$

$$AW_5 = -70000(0.263) - 20000 + 10000(0.16) \\ = \$-36600$$

Chapter no 13

Trade In:

A trade-in when you give your old equipment, machine, or item to a seller in exchange for money towards the purchase of new one.

Important Note:

In replacement studies, the trade-in value is not always equal to fair market value. It might be less because dealers can offer lower prices. So it shouldn't be use unless it reflects the true market value.
→ yeh pata nahi meine kyon likha ha ;

$$* P - (TEV - MV)$$

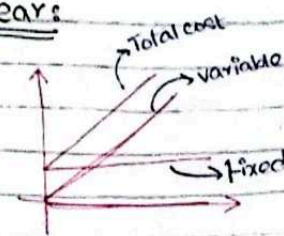
↓ ↓ ↓
 Challenger's Trade in Defender's 1st
 1st cost value cost

Steps:

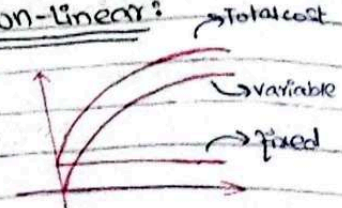
- Retailer
- Book value
- Own opinion

yeh mujhe pata he nahi chala chpt 13 kab start hua

Linear:



Non-linear:



$$* \text{Profit} = \text{Revenue} - \text{Total cost} \\ = R - TC \\ = R - (FC + VC) \\ \downarrow \quad \downarrow \\ \text{fixed} \quad \text{variable}$$

$$* R = TC \quad (\text{Revenue a Variable cost}) \\ \text{Revenue} \leftarrow RQ = FC + VQ \\ Q = \frac{FC}{R - V} \quad Q = \frac{F}{r - v}$$

Chapter no 13

Break even and Payback Analysis

Example # 13.1

Fixed cost = \$2,400,000/month
 Variable cost per unit (V) = \$35,000
 Revenue (r) = \$75,000
 Normal production = 60 units/month
 Current = 72 units/month

a) Breakeven point units:

$$\begin{aligned}\text{Breakeven} &= \frac{\text{Fixed}}{\text{Revenue} - \text{Variable}} = \frac{FC}{r-v} \\ &= \frac{2,400,000}{75,000 - 35,000} = 60 \text{ units}\end{aligned}$$

→ Breakeven is 60, while company is making 72, so company is at profit.

b) Current profit at 72:

$$\begin{aligned}\text{Profit} &= \text{Total Revenue} - \text{Total cost} \\ \text{Total Revenue} &= 72 \times 75,000 = 5,400,000 \\ \text{Variable cost} &= 72 \times 35,000 = 2,520,000 \\ \text{Total cost} &= \text{Fixed} + \text{variable} \\ &= 2,400,000 + 2,520,000 \\ &= 4,920,000 \\ \text{Profit} &= 5,400,000 - 4,920,000 \\ &= \$480,000\end{aligned}$$

Alternative:

$$\begin{aligned}\text{Profit} &= (r-v)Q - FC \\ &= (75-35)72 - 2400 \\ &= \$480\end{aligned}$$

c) 45 units:

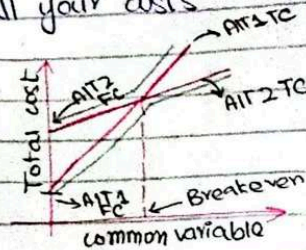
$$r-v = \frac{FC}{Q}$$

$$Q = 45$$

$$r-v = \frac{2,400,000}{45} = \$53,333.33$$

Breakeven between two analysis:

→ the number you must sell to cover all your costs — No profit, No Loss



→ Alt 1 has Lower fixed cost but higher variable cost

→ Alt 2 has higher fixed cost but Lower variable cost

Decision:

- If production < Breakeven, choose Lower FC
- If production > Breakeven, choose Lower VC

Example # 13.2

Let x = tone per year
 $i = 10\%$

Auto Feed Machine:

First cost = \$23,000, Salvage = \$4,000
Life = 10 years, Annual M&O = \$3,500
Labor = 1 person at \$12/hour
Output = 8 tons/hour

$$\text{Labor cost} = \frac{x \text{ tons}}{8 \text{ tons/hour}} = \frac{x}{8} \text{ hour}$$
$$= \frac{x}{8} \cdot 12 = 1.5x$$

$$\text{Total AC} = -23000(A/P, 10\%, 10) + 4000(A/F, 10\%, 10)$$
$$- 3500 - 1.5x$$
$$= \$ -6992 - 1.5x \quad \text{--- (1)}$$

Manual Feed Machine:

First cost = \$8,000, Salvage = 0
Life = 5 years, Annual M&O = \$1,500
Labor = 3 workers at \$8/hr = \$24/hr
Output = 6 tons/hour

$$\text{Labor cost} = \frac{x}{6} \cdot 24 = 4x$$

$$\text{Total AC} = -8000(A/P, 10\%, 5) - 1500 - 4x$$
$$= -3610 - 4x \quad \text{--- (2)}$$

Now,

$$-6992 - 1.5x = -3610 - 4x$$

$$x = 1353 \text{ tons/year}$$

This is breakeven rate, auto machine becomes a better option.

Examples

Initial cost = \$12000, Life = 3 years
Salvage = \$2000, Employer = \$45000/year
In addition, each ad = \$8/delivery
Total = 4000 ads

$$\text{Alt 2} = \$20/\text{ad} \quad i = 8\%$$

Breakeven amount = ?

$$-12000(A/P, 8\%, 3) - 45000 + 2000(A/F, 8\%, 3)$$
$$- 8x = -20x$$

$$-12000(0.560) - 45000 + 2000(0.308) - 8x = 20x$$

$$x = 4087 \text{ ads/year}$$

konsa alt
choose karo
tha ye mei
choot gayi hu

Chapter no 16

Depreciation Methods

What is it?

Depreciation is a way for businesses to show that something they own, like machine or vehicle, is losing value over time. It is a non-cash expenses used in accounting to spread the cost of the asset over its useful life. It doesn't mean money is spent, it just shows that the asset is getting older or used up.

→ Depreciation amount = D

→ Depreciation rate = d

1 - Straight line depreciation:

- Most commonly used; assumes equal expense each year.

Formula:

$$Dt = \frac{\text{Cost} - \text{Salvage value}}{\text{Useful life}} = \frac{B - S}{n}$$

Dt = Annual depreciation charged

	$BV_t = BV_{t-1} - D_t$
	EXAMPLE # 16
	$n = 12$, $DPDB \Rightarrow 2/n$
	First cost = \$ 25000
	Salvage value = \$ 2500
	AZHAR BOOK

Date: _____	Mo Tu We Th Fr Sa Su /
depreciation amount = ? } for 4 th year	
Book value = ? }	
$D_t = (d)/BV_{t-1}$	$DDP = \frac{2}{n} = \frac{2}{12}$
$= (2/12) = 0.1667$	$= 0.1667$
$\Rightarrow D_1 = (0.1667)(25000) = \4166	
$BV_1 = 25000 - 4166$	
$= 20834$	
$D_2 = (0.1667)(25000)$	
$= \$4166$	
$BV_2 = (25000)(1 - 0.1667) = 20833$	
$D_4 = 2411$	
$BV_4 = 12054$	
depreciation amount	Book value
$D_1 = 4166$	$BV_1 = 20834$
$D_2 = 3473$	$BV_2 = 17361$
$D_3 = 2894$	$BV_3 = 14467$
$D_4 = 2411$	$BV_4 = 12056$
$D_5 = 2009$	$BV_5 = 10047$
$D_6 = 1674$	$BV_6 = 8373$
$D_7 = 1395$	$BV_7 = 6978$
$D_8 = 1163$	$BV_8 = 5814$
$D_9 = 969$	$BV_9 = 4845$
$D_{10} = 807$	$BV_{10} = 4038$
	AZHAR BOOK

Date: _____	Mo Tu We Th Fr Sa Su
$D_{11} = 673$	$BV_{11} = 3365$
$D_{12} = 560$	$BV_{12} = 2805$
29/4 * Previous Question without formula	
$\rightarrow B = 25,000$, $S = 2500$, $n = 12$, $d = 2/12 = 0.1667$	

2 - Declining Balance:

- Accelerated method; higher depreciation in earlier years
- Double Declining balance (DDB) is most popular

Formula:

$$\text{Depreciation Expense} = 2 \times \frac{\text{Straight Line}}{\text{Line}} \times \text{Book value at start of year}$$

$$\begin{array}{ccc} * d_{\max} = \frac{2}{n} & ; & d = \frac{1.5}{n} \\ \downarrow & & \downarrow \\ \text{Declining blnc} & & \text{Double declining blnc} \end{array}$$

Example # 16.2 (Declining balance)

$$B = 25000 ; S = 2500 , n = 12$$

N	Balance	Dep(d)	Remain	Dep(D)	Book value
1	100	16.67	83.33	4167.5	20832.5
2	83.33	13.89	69.44	3472.5	17360
3	69.44	11.57	57.86	2892.5	14465
4	57.86	9.64	48.22	24100.1	12055
5	48.22	8.03	40.19	2007.5	10471 10047.5
6	40.19	6.69	33.5	1672.5	8375
7	33.5	5.58	27.92	1395.0	6980
8	27.92	4.65	23.27	1162.5	5817.5
9	23.27	3.87	19.4	967.5	4850
10	19.4	3.23	16.16	807.5	4040
11	16.16	2.69	13.46	672.5	3365
12	13.46	2.24	11.21	560	2802.5

Step I: Always start balance with 100

Step II: $d = \frac{2}{n} = \frac{2}{12} = 0.1667 = 16.67\%$
put in d

Step III: Balance - d = Remain
 $100 - 16.67 = 83.33$

Step IV: Remain value in Balance (value no 2)

Step V: $83.33 \times 16.67\% = 13.89$ (d value no 2)
and then again step 3
After filling first three column

Step VI: $\text{Dep(D)} = B \times d\%$
 $= 25000 \times 16.67\% = 4167.5$
 $= 25000 \times 13.89\% = 3472.5$

Step VII: $B.V = B \times \text{Remain}(\%)$
 $= 25000 \times 83.33 = 20832.5$
 $= 25000 \times 69.44 = 17360$

Example:

$$B = 25000 ; S = 2500 ; n = 7 \text{ years}$$

$$D.B = \frac{1.5}{7} = 0.214 \rightarrow \frac{0.214}{2} = 0.107$$

N	Balance	d	Remain	N*	Straight	MACRS	D	BV
1	100	10.7	89.3	—	—	10.7	2675	22325
2	89.3	19.11	70.18	6.5	13.73	19.11	4777.5	17547.5
3	70.18	15.01	55.16	5.5	12.76	15.01	3752.5	13795
4	55.16	11.80	43.35	4.5	12.25	12.25	3062.0	10732.5
5	43.35	9.27	34.07	3.5	12.25	12.25	3062.0	7670
6	34.07	7.29	26.77	2.5	12.25	12.25	3062.0	4607.5
7	26.77	5.72	21.04	1.5	12.25	12.25	3062.0	1545
8	21.04	21.04	0	0.5	12.25	12.25	1531	13.5
								24986.5

→ Balance, d, Remain → same method

→ N* → first blank → then $7.5 - 1 = 6.5$ → keep question subtracting 1

→ Straight line → first blank → $\frac{\text{Balance}}{N^*} = \frac{89.3}{6.5} = 13.73$

until straight $\geq d$, then we use that only value
i.e. $12.25 > 11.80$

→ MACRS = d until straight > d then
MACRS = straight
(till last value)

$$\rightarrow D = \text{MACRS}^{(\%) \times B}$$

$$~~(1)10.7 \times 100 = 10.7\% \times 25000 = 2675~~$$

$$19.11\% \times 25000 = 4777.5$$

$$\rightarrow BV = 25000 - D \text{ and so on}$$

$$= 25000 - 2675 = 22325$$

$$= 22325 - 4777.5 = 17547.5$$

